LABORATORY-SCALE PLANETARY SURFACE MAPPING USING USRP-BASED SYNTHETIC APERTURE RADAR

Report submitted to the SASTRA Deemed to be University in partial fulfillment of the requirements for the award of the degree of

B. Tech. Electronics & Communication Engineering

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SCHOOL OF ELECTRICAL & ELECTRONICS ENGINEERING

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This is to certify that the report titled **"Laboratory-Scale Planetary Surface Mapping Using USRP-Based Synthetic Aperture RADAR"** submitted in partial fulfillment of the requirements for the award of the degree of B. Tech. **Electronics & Communication Engineering** to the SASTRA Deemed to be University, is a bona-fide record of the work done by **Mr. SUDEEPAN V** (Reg. No.: 127004266), **Mr. MUGESH C** (Reg. No.: 127004162) and **Mr. GIRINATH N U** (Reg. No.: 127004076) during the Sixth semester of the academic year 2025-26, in the **School of Electrical & Electronics Engineering**, under my supervision. This report has not formed the basis for the award of any degree, diploma, associateship, fellowship or other similar title to any candidate of any University.

Signature of Project Supervisor :

Name with Affiliation : Dr. James A Baskaradas, Professor

Date :

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Declaration

We declare that the report titled **"Laboratory-Scale Planetary Surface Mapping Using USRP-Based Synthetic Aperture RADAR"** submitted by us is an original work done by us under the guidance of **Dr. James A Baskaradas, Professor**, **School of Electrical and Electronics Engineering, SASTRA Deemed to be University** during the Sixth semester of the academic year **2025-26**, in the **School of Electrical and Electronics Engineering**. The work is original and wherever we have used materials from other sources, we have given due credit and cited them in the text of the report. This report has not formed the basis for the award of any degree, diploma, associate-ship, fellowship or other similar title to any candidate of any University.

Signature of the candidates:

Sudeepan V Mugesh C Girinath N U

Date:

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ABSTRACT

Planetary surface mapping is an important area in remote sensing and space exploration. It helps scientists understand the geology, shape, and physical makeup of planets. Traditional optical and imaging mapping methods have notable limitations. They depend on sunlight, are sensitive to the atmosphere, and cannot see through surface layers. Synthetic Aperture Radar (SAR) is an active microwave sensor that solves these problems. It can capture images in any weather and at any time, regardless of lighting. Yet, current spaceborne and airborne SAR setups require expensive, specialized equipment, making it hard to demonstrate in labs for education and research.

This project details the design, implementation, and evaluation of a lab-scale planetary surface mapping system. It uses a Universal Software Radio Peripheral (USRP) B210 software defined radio platform, controlled by GNU Radio, with a wideband double ridged horn antenna that operates from 1 GHz to 8 GHz. The system was first set up using Frequency Modulated Continuous Wave (FMCW) SAR at a center frequency of 6 GHz with a 5 MHz bandwidth and a sample rate of 5 MSPS. Data was gathered from 64 positions with a step size of 15 mm at a target distance of about 2.5 meters. The FMCW method proved unreliable because of sampling errors, bandwidth limits, and not enough samples per chirp, which resulted in low resolution of the final SAR image. Therefore, the system was redesigned to use Stepped Frequency Continuous Wave (SFCW) radar, which sweeps through 501 distinct frequency steps from 1 GHz to 6 GHz. This provides a total bandwidth of 5 GHz and a theoretical range resolution of 3 cm. The SFCW method showed much better signal stability, accurate measurements for each frequency step, and dependable SAR matrix formation. Signal processing in MATLAB with a backprojection algorithm produced a 2D SAR image that displayed range versus cross-range information of the scene.

Keywords: Synthetic Aperture Radar (SAR), USRP B210, FMCW, SFCW, GNU Radio, Software Defined Radio (SDR), Back-Projection Algorithm, Planetary Surface Mapping, IQ Sampling, SAR Matrix.

Specific Contribution and Learning — Girinath N U (127004076)

Specific Contribution: Designed and implemented the FMCW GNU Radio flowgraph (fmcw_sar_robust.py) including chirp generation, multiply-conjugate mixing, low-pass filtering, and file sink configuration. Led the hardware setup and antenna alignment experiments.

Specific Learning: Gained practical understanding of FMCW radar signal chain, IQ sampling, GNU Radio block-level programming, and the challenges of real-time SDR streaming with the USRP B210 hardware.

Specific Contribution and Learning — Mugesh C (127004162)

Specific Contribution: Developed the automated aperture data collection script (run_sar.py) that manages multi-position capture, file validation, retry logic, and SAR matrix assembly. Assisted in interpreting FMCW results and identifying hardware bottlenecks.

Specific Learning: Gained expertise in Python-based automation of SDR hardware, subprocess management, binary IQ data parsing, and the process of assembling raw radar data into structured SAR matrices.

Specific Contribution and Learning — Sudeepan V (127004266)

Specific Contribution: Designed and implemented the SFCW radar system (sfcw_sar.py) including frequency-stepped UHD direct API control, complex IQ measurement per frequency step, SAR matrix construction, and MATLAB .mat file export. Led the MATLAB back-projection SAR image formation.

Specific Learning: Developed deep understanding of SFCW radar principles, UHD Python API, frequency tuning settle time management, IFFT-based range profiling, and SAR back-projection image formation algorithm.

Signature of the Guide Student Reg. No: 127004266 / 127004162 / 127004076

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ABBREVIATIONS

Abbreviation	Full Form
SAR	Synthetic Aperture Radar
FMCW	Frequency Modulated Continuous Wave
SFCW	Stepped Frequency Continuous Wave
USRP	Universal Software Radio Peripheral
SDR	Software Defined Radio
GRC	GNU Radio Companion
UHD	USRP Hardware Drive
IQ	In-phase and Quadrature
LPF	Low Pass Filter
FFT	Fast Fourier Transform

Abbreviation	Full Form
BW	Band Width
TX	Transmitter
RX	Receiver
SNR	Signal-to-Noise Ratio
LFM	Linear Frequency Modulation
RMA	Range Migration Algorithm
RCM	Range Cell Migration
RCMC	Range Cell Migration Correction
DFT	Discrete Fourier Transform
GPS	Global Positioning System
ESA	European Space Agency
GBSAR	Ground Based Synthetic Aperture Radar
IFFT	Inverse Fast Fourier Transform
MSPS	Mega samples Per Sec
ADC	Analog-to-Digital Converter
RF	Radio Frequency

NOTATIONS

English Symbols (in alphabetical order):

Symbol	Description	Unit
BW	Total bandwidth of the radar system	Hz
c	Speed of light	m/s
D	Largest physical dimension of antenna aperture	m

Symbol	Description	Unit
df	Frequency step in FMCW	Hz
f_0	Carrier/center frequency	Hz
f_b	Beat frequency in FMCW	Hz
K	Chirp rate (sweep rate) in FMCW	Hz/s
L	Antenna length in flight direction	m
N	Number of frequency steps in FMCW	-
R	Range (target distance)	m
S(x,t)	SAR raw data matrix (FMCW) - position vs time	-
S(x,f)	SAR raw data matrix (SFCW) - position vs frequency	-
Ts	Chirp sweep time	s
W	Ground swath width	m
x	Azimuth (cross range) antenna position	m

Greek Symbols (in alphabetical order):

Symbol	Description	Unit
δ_{az}	SAR azimuth (cross-range) resolution	m
δ_r	Range resolution	m
λ	Wavelength at center frequency	m
θ_{az}	Azimuth beamwidth of the antenna	degrees
θ_{el}	Elevation beamwidth of the antenna	degrees
ρ_{az}	Azimuth resolution of the synthetic aperture	m

CHAPTER 1 : INTRODUCTION

1.1 OVERVIEW OF PLANETARY SURFACE MAPPING

Mapping the surfaces of other planets is one of the more humbling exercises in modern science. It demands that we understand distant, alien worlds well enough to interpret what we see, while working with instruments that must survive launch, space travel, and extreme environmental conditions. The scientific stakes are high: accurate surface characterization tells us about geological history, helps identify landing sites that won't swallow a rover in loose regolith, and sometimes hints at the presence of resources or even conditions hospitable to life. Whether it is the cratered highlands of the Moon, the layered polar deposits of Mars, or the hydrocarbon lakes of Titan, the ability to produce reliable maps is foundational to almost every aspect of planetary science.

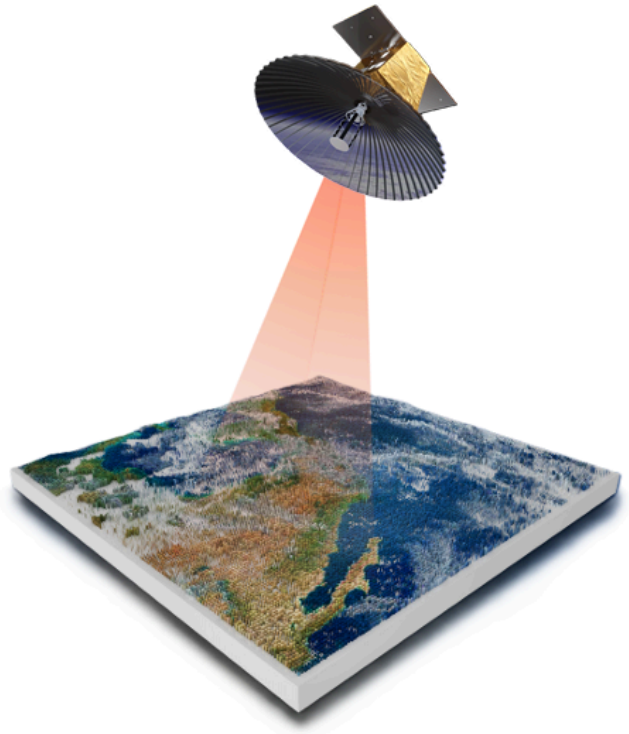


Fig.1.1 Satellite Scanning the earth surface using SAR

For much of the space age, visible-light cameras and infrared instruments were the primary tools for this job. They work well enough when the sun is shining and the atmosphere cooperates but that is a significant caveat. Venus, perpetually wrapped in a

thick sulphuric acid cloud deck, is simply invisible to optical cameras. Mars occasionally goes dark for months beneath global dust storms. Even on bodies with no real atmosphere, features buried under dust or ice layers remain completely opaque to optical wavelengths. These are not edge cases; they represent some of the most scientifically interesting environments in the solar system.

Microwave radar sidesteps all of these problems at once. Because radar generates its own signal rather than relying on reflected sunlight, it works in complete darkness. Because centimetre to decimetre wavelengths pass through clouds and dust without significant attenuation, surface coverage is maintained regardless of atmospheric conditions. And at certain frequencies, radar signals can even penetrate shallow surface layers, revealing buried structures that would otherwise remain invisible. These properties have made radar the sensor of choice wherever optical systems fall short.

The development of Synthetic Aperture Radar added another dimension to this capability. Rather than being limited by the physical size of its antenna which determines resolution in a conventional radar system SAR synthesises the equivalent of a very large aperture by coherently combining echoes gathered as the platform moves along its flight path. The result is fine spatial resolution that, crucially, does not degrade with distance. SAR systems have mapped Venus (Magellan), contributed to Mars subsurface studies, and form the backbone of Earth observation constellations like Sentinel-1. This project takes SAR into a very different setting: a university laboratory to ask whether the same basic principles can be demonstrated at low cost using commercially available radio hardware.

1.2 Why RADAR Over Optical Systems

The case for radar over optical imaging in planetary contexts really comes down to a handful of physical facts. The first and most obvious is independence from solar illumination. A radar system carries its own energy source and does not need the sun. It can image a permanently shadowed crater at a lunar pole just as easily as a sunlit equatorial plain. Optical systems cannot.

The second advantage is atmospheric penetration. At the microwave wavelengths typically used in radar, clouds, aerosols, and dust are essentially transparent. This is what allowed the Magellan spacecraft to produce the first global map of Venus despite that planet's impenetrable cloud cover, something no optical camera has ever managed. On Mars, radar observations continue even through the dust storm seasons that periodically make the planet optically opaque.

Third, certain radar frequencies particularly in the lower microwave bands can partially penetrate the surface itself. Ground penetrating radar on Mars orbiters has revealed subsurface layering and is thought to have detected evidence of liquid water beneath the southern polar cap. This subsurface sensitivity is simply not available to any passive optical sensor.



Fig.1.2 Synthetic Aperture Radar (SAR) Image overlaid on Optical Map

Finally, radar is a coherent sensor: it preserves both the amplitude and phase of the received signal. This opens up an entirely different class of analysis techniques unavailable with optical cameras. Interferometric SAR, for example, can detect surface deformation at millimetre scales by comparing the phase of two images taken from slightly different positions or at different times. Polarimetric SAR can extract information about the physical properties and orientation of surface materials. These

techniques have made radar instruments scientifically productive far beyond their basic imaging function which is precisely why investing in low-cost radar education matters.

1.3 Synthetic Aperture Radar Concept

The central challenge in radar imaging is achieving fine resolution in the direction perpendicular to the radar beam the azimuth, or cross-range, direction. In a conventional real-aperture radar, this resolution is set by the antenna beamwidth: a narrow beam gives fine resolution, but a narrow beam requires a physically large antenna. At the ranges typical of satellite remote sensing, the antenna would need to be kilometres long to achieve useful resolution, obviously impractical.

SAR's solution is elegant. Instead of building a larger antenna, it uses the platform's motion to synthesize one. As the platform travels along its flight path, the radar continuously transmits and receives pulses. Because any given target on the ground remains within the antenna beam for an extended period spanning many platform positions its echoes are recorded from multiple slightly different angles. Coherent processing of all these echoes across the full extent of the synthetic aperture collapses the returns from that target into a single focused image pixel. The synthetic aperture length, and thus the achievable cross-range resolution, is determined by how long the target stays within the beam which is in turn set by the real antenna's beamwidth.

This leads to the famous result that azimuth resolution equals approximately half the physical antenna length, regardless of range:

$$\delta_{az} = L/2 \tag{1}$$

Counter-intuitively, this means a smaller antenna gives better resolution, since its wider beam keeps targets illuminated for longer, producing a longer synthetic aperture. Range resolution, by contrast, is determined entirely by the signal bandwidth:

$$\delta_{r} = \frac{c}{(2\times BW)} \tag{2}$$

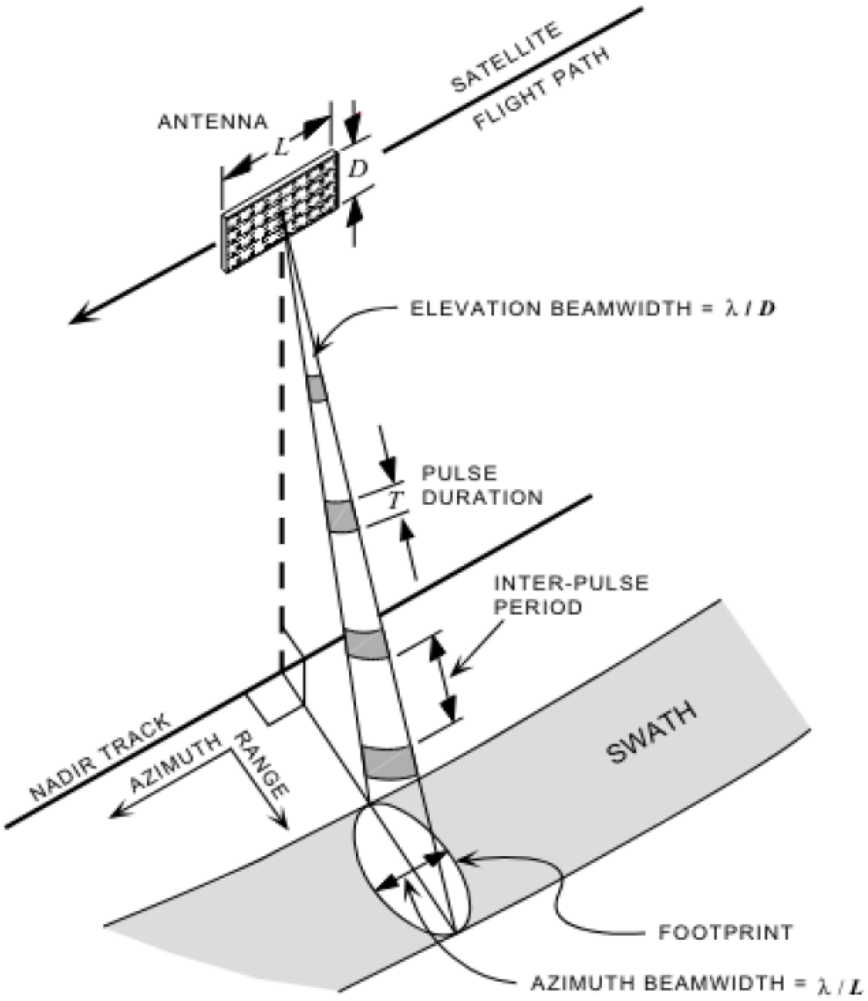


Fig.1.3 Geometry of Synthetic Aperture Radar (SAR) System

where c is the speed of light and BW is the transmitted bandwidth. For our SFCW system spanning 5 GHz from 1 to 6 GHz, this works out to a theoretical range resolution of 3 cm more than adequate for identifying objects at the laboratory scale. SAR image formation then requires a two-dimensional processing operation: range compression in the fast-time direction (via FFT or IFFT), followed by coherent aperture synthesis in the slow-time (azimuth) direction using algorithms such as back-projection, the Range-Doppler Algorithm, or the Range Migration Algorithm.

1.4 Problem Statement

Optical and imaging sensors still dominate most planetary surface mapping payloads, but their dependence on sunlight and atmospheric transparency means there will always be targets and conditions where they fail completely. Radar and SAR specifically addresses all of these shortcomings. The problem is that real SAR systems are expensive. The hardware required for a spaceborne SAR mission runs into hundreds of millions of dollars, and even airborne systems cost millions. This pricing structure puts direct hands-on experience with SAR data collection entirely out of reach for most university laboratories.

This project addresses that gap directly. The challenge we set ourselves was to design, build, and evaluate a laboratory-scale SAR system using the USRP B210 software-defined radio, a platform that costs a fraction of dedicated radar hardware and to determine whether such a system can produce a meaningful focused 2D SAR image of a laboratory scene. The goal was not to replicate a spaceborne system, but to demonstrate that the fundamental principles of SAR data collection and image formation are accessible at the bench scale with commodity hardware.

1.5 Scope of the Project

Within this framing, the project covered the following activities: designing and implementing an FMCW SAR data collection system using GNU Radio and the USRP B210; building an automated aperture scanning script to manage 64 position data collection; investigating and documenting the limitations of FMCW on this hardware; redesigning the system using the SFCW approach via the UHD Python API; collecting SFCW SAR data and assembling the SAR matrix; forming a 2D SAR image in MATLAB using back-projection; and drawing a systematic comparison between the two waveform strategies. The project deliberately stays within the laboratory environment and does not attempt to simulate free-space or far-field planetary conditions, which remain topics for future work.

CHAPTER 2 : LITERATURE REVIEW

S No.	Title	Journal	Interferance
1	Design and Implementation of Compact 77 GHz Synthetic Aperture Radar for Drone Based Applications	2019 6th Asia-Pacific conference on Synthetic Aperture Radar (APSAR)	The paper shows the design of a compact 77 GHz millimeter-wave SAR system integrated with a UAV (drone) to provide lightweight and efficient airborne radar imaging. Experimental results indicate that the proposed system can effectively acquire SAR data and produce radar images while being small, lightweight, and suitable for drone-based remote sensing applications.
2	FMCW SAR Imaging Based on Software Designed Radio Architecture	2023 International Telecommunications Conference (ITC-Egypt)	The paper discusses an FMCW Synthetic Aperture Radar (SAR) imaging system built using a Software Defined Radio (SDR) architecture. This setup allows for flexible signal generation, acquisition, and processing. Experimental findings reveal that the SDR-based FMCW SAR system can produce SAR images at a lower cost, with high flexibility and easy reconfiguration compared to traditional radar hardware systems.
3	SAR Image Stitching Method for MxN SAR sub-images	IEEE 2024	The paper introduces a SAR image stitching method. This method combines phase correlation-based sub-image registration with least-squares global optimization to accurately align multiple SAR sub-images and determine their global positions. The approach also uses gray-scale equalization and seam blending to eliminate intensity discontinuities, resulting in a large-scene SAR image with a better field of view and higher resolution.
4	Utilizing Polarization Diversity in GBSAR Data-Based Object	Sensors 2024	The paper presents a Ground-Based SAR (GBSAR) system that classifies close-range objects based on their

S No.	Title	Journal	Interferance
	Classification		shape and material composition, such as metal, glass, or plastic. It does this by analyzing their unique dielectric properties and radar cross-sections. Experimental results show that by using localized scanning and polarization diversity, the system can effectively differentiate targets of the same size but different materials. This demonstrates the potential of small-scale, ground-based setups for high-resolution physical classification.
5	Fast-Fourier Time-Domain SAR Reconstruction for Millimeter-Wave FMCW 3-D Imaging	IEEE Transactions on Microwave Theory and Techniques (Volume: 72, Issue:12, December 2024)	The paper describes a precise phase calibration technique and a highly efficient time-domain reconstruction algorithm specifically designed for 3-D millimeter-wave FMCW SAR imaging. Experimental results indicate that the proposed holographic algorithm can create high-resolution 3-D images of targets and concealed objects at close range. It operates two orders of magnitude faster than current time-domain methods, making it very effective for near-field, localized radar setups like the USRP configuration.

CHAPTER 3 : METHODOLOGY

3.1 System Overview

The complete system we built and used for this project can be broken into three parts that had to work together: the hardware platform, the data acquisition software, and the signal processing pipeline. On the hardware side, the USRP B210 transceiver connects via USB 3.0 to a laptop running Ubuntu Linux, with the double ridged horn antenna serving as the single combined transmit and receive element. The data acquisition layer consists of two distinct implementations the initial GNU Radio-based FMCW flowgraph (fmcw_sar_robust.py) and the later UHD-direct SFCW script (sfcw_sar.py), both coordinated by an automation script (run_sar.py) that handles the multi-position collection loop. Signal processing then happens offline in MATLAB, which performs background clutter subtraction, windowing, IFFT-based range profile formation, and the back-projection algorithm that assembles the final 2D SAR image.

A typical data collection session works as follows. The antenna is mounted on a tripod at roughly two metres above the floor, aimed at the target scene approximately 2.5 metres away. We defined a linear aperture in the horizontal direction and moved the antenna manually between positions in 15 mm steps, for a total of 64 positions spanning 94.5 cm. At each position, we ran the data collection script and stored the received IQ data to disk. After all 64 positions were captured, the per-position files were assembled into a single 2D SAR data matrix and saved in MATLAB format for the processing stage.

3.2 Hardware Description

3.2.1 USRP B210 Software Defined Radio

The USRP B210 from Ettus Research is a full-duplex SDR transceiver built around the Analog Devices AD9361 RF front-end chip. It covers 70 MHz to 6 GHz, supports instantaneous bandwidths up to 56 MHz per channel, and connects to the host computer via USB 3.0. In practice, stable streaming without dropped samples tends to cap out at around 20 MSPS in typical environments; we ran ours at 5 MSPS to be safe.

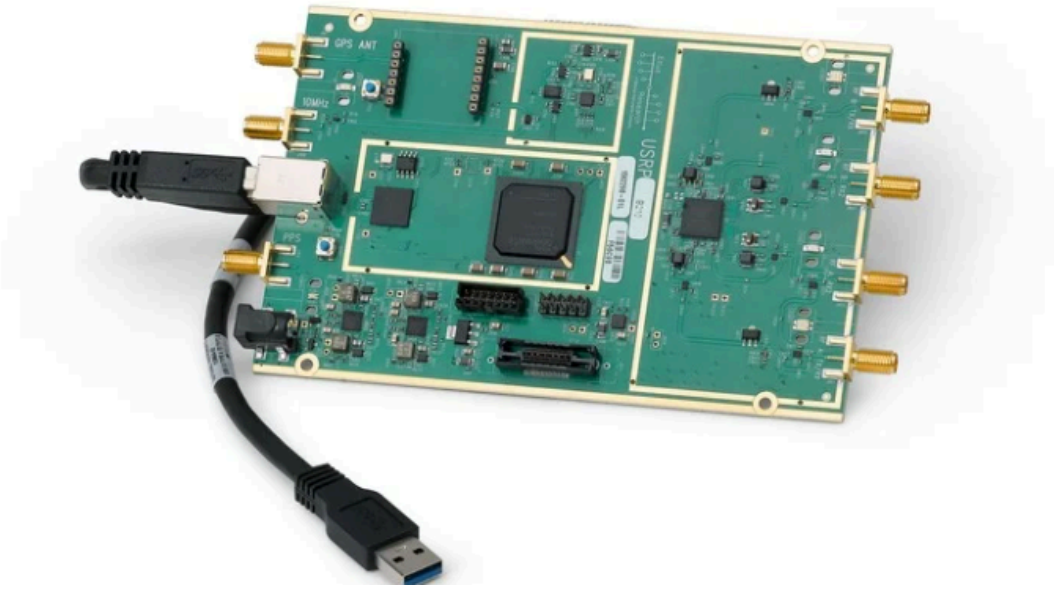


Fig.3.1 USRP B210 software-defined radio used for transmitting and receiving radar signals in the SAR system.

We configured the B210 in a monostatic arrangement with transmit on the TX/RX port and receive on RX2, both on Channel A. The master clock rate was set to 40 MHz, which is a well-known stable operating point for this device. For the FMCW phase, both channels were fixed at 6 GHz. For SFCW, we stepped the center frequency through 501 values from 1 GHz to 6 GHz. Hardware control went through the Ettus Universal Hardware Driver (UHD) via GNU Radio's UHD blocks in the FMCW case, and directly via the UHD Python API (libpyuhd) for the SFCW implementation.

3.2.2 Horn Antenna

The antenna is a commercial wideband double-ridged horn covering 1 to 8 GHz, designed primarily for EMC testing applications. The double-ridged horn geometry achieves broad bandwidth by loading a standard pyramidal horn with two parallel metallic ridges in the E-plane, which suppresses the cutoff of the dominant mode while keeping efficiency high. The physical aperture is 241 mm wide and 153 mm tall. At 2 GHz, the gain is 10.66 dBi, the azimuth half-power beamwidth is approximately 43.5° and the elevation beamwidth is about 75.5°.

$$\theta = \frac{70\lambda}{D} \tag{3}$$

$$\rho_{az} = \frac{L}{2} = \frac{0.153}{2} = 0.0765\, m \approx 7.65\, cm \tag{4}$$

The SAR azimuth resolution set by the antenna's 153 mm dimension is:

Table 3.1 Antenna Specifications — Double Ridged Horn

Parameter	**Value**
Operating frequency range	1 GHz – 8 GHz
Physical aperture width (D)	241 mm
Physical aperture height	153 mm
Physical aperture area (Ap)	0.037 m²
Effective aperture (Ae at 2 GHz)	0.0209 m²
Antenna gain at 2 GHz	10.66 dBi
Azimuth beamwidth (2 GHz)	43.5°
Elevation beamwidth (2 GHz)	75.5°
SAR azimuth resolution	7.65 cm
Ground swath width (at 2 m height	1.6 m
Ground swath length	3.09 m

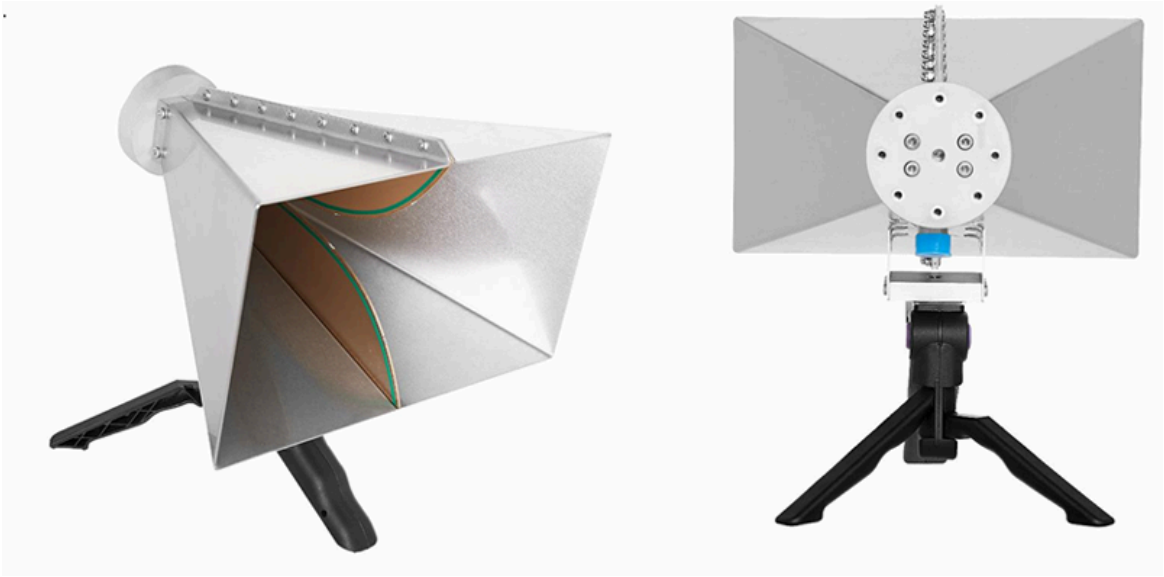


Fig.3.2 Double-ridged horn antenna used for transmitting and receiving microwave signals in the SAR system.

3.3 FMCW SAR Implementation

3.3.1 FMCW Operating Principle

In FMCW radar, the transmitted signal is a continuous wave whose frequency is linearly swept chirped across a defined bandwidth in each measurement interval. Unlike pulsed radar, FMCW transmits and receives simultaneously. Range information comes not from measuring a time-of-flight delay directly, but from the frequency difference between the transmitted and received signals at any instant. The transmitted waveform is:

$$s_{tx}(t) = A\cos(2\pi f_0 t + \pi K t^2) \quad (5)$$

where f_0 is the starting frequency and K = BW/Ts is the chirp rate. The echo from a target at range R arrives delayed by $\tau = 2R/c$, and when the received signal is mixed with a copy of the transmitted signal and low-pass filtered, the result is a beat signal at frequency:

$$s_{rx}(t) = A\cos(2\pi f_0(t - \tau) + \pi K(t - \tau)^2) \quad (6)$$

$$f_b = K \times \tau = K \times (\frac{2R}{c}) = \frac{2KR}{c} \quad (7)$$

$$R = \frac{(c \times f_b)}{2K} \quad (8)$$

Range is then recovered simply by reading off the beat frequency. The Fourier transform of the beat signal gives a range profile in which each peak corresponds to a target at a specific distance.

3.3.2 GNU Radio Implementation

The FMCW system was implemented as a GNU Radio flowgraph in fmcw_sar_robust.py. A sawtooth signal source at 1 kHz sweeps the instantaneous frequency between -2.5 MHz and +2.5 MHz over each 1 ms chirp period. A frequency modulator block converts this to a complex IQ waveform representing the chirp. The chirp drives the USRP transmitter (after a 0.7 amplitude scaling) and simultaneously feeds the first input of a multiply-conjugate block; the USRP receiver feeds the second input. The multiply-conjugate output is the dechirped beat signal, which then passes through a Hamming windowed FIR low pass filter with a 2 kHz cutoff. A skip-head block discards the first 1.5 million samples (300 ms at 5 MSPS) to let the hardware settle, and a head block then captures exactly 160,000 samples corresponding to 32 complete chirp periods before writing them to disk.

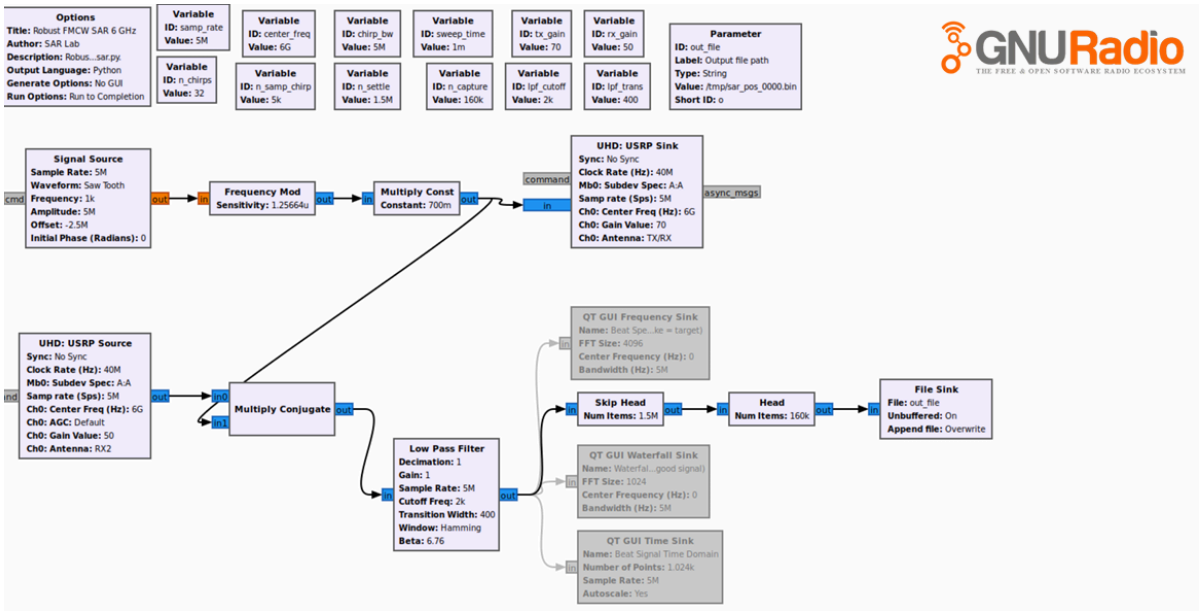


Fig.3.3 GNU Radio flowgraph for FMCW SAR signal generation, mixing, filtering, and data acquisition using USRP B210

Table 3.2: FMCW SAR System Parameters

Parameter	Symbol	Value
Center frequency	f_0	6 GHz
Chirp bandwidth	BW	5 MHz
Sample rate	fs	5 MSPS
Sweep time	Ts	1 ms
Samples per chirp	n_samp_chirp	5000
Chirps per position	n_chirps	32
Total capture per position	n_capture	1,60,000 Samples
Settling skip	—	300 ms (1.5 M samples)

Parameter	Symbol	Value
TX gain	GTX	70 dB
RX gain	GRX	50 dB
LPF cutoff	fc	2 KHz
Aperture positions	N	64
Step size	dx	15 mm
Target distance	R	~2.5 m
Theoretical range resolution	δr	30 m (BW = 5MHz)

3.4 Data Acquisition Process

3.4.1 run_sar.py Automation Script

Manually launching the GNU Radio flowgraph at each of 64 positions and keeping track of files would have been tedious and error-prone. We wrote run_sar.py to handle the entire collection campaign automatically. At startup, the script computes and displays all key system parameters including derived quantities like range and cross-range resolution and saves them to a JSON configuration file alongside the data, so that the MATLAB processing stage has everything it needs without requiring any manual parameter entry.

The main loop iterates over all 64 positions, prompts the operator to physically move the antenna to the next location, then calls a function that launches fmcw_sar_robust.py as a subprocess. One persistent issue in our WSL2 environment was that the GNU Radio process sometimes refused to exit cleanly after capturing its data (the UHD hardware teardown would hang). We handled this with a two-level timeout: wait 10 seconds for normal exit; if that fails, send SIGINT to flush the buffer; if the process still hasn't died

after another 3 seconds, send SIGKILL. This approach reliably recovers the captured data in nearly all cases.

After each capture, a validation function checks that the output file exists, has exactly the expected size (1,280,000 bytes for 160,000 complex float32 samples), and that the first few samples contain finite, reasonable values. Failed validations trigger an automatic retry, up to three attempts. Once all 64 positions are collected, the per-position binary files are loaded, reassembled into a 64 × 160,000 complex matrix, and saved as a .npy file alongside the configuration JSON.



Fig.3.4 Experimental SAR data acquisition setup showing horn antennas, USRP module, and measurement arrangement in the laboratory environment.

3.4.2 IQ Sampling and SAR Matrix Formation

At the hardware level, the AD9361 chip inside the B210 digitises the received signal at 5 MSPS and outputs complex IQ samples (one in-phase and one quadrature component)

per sample interval. The complex baseband representation s[n] = I[n] + jQ[n] preserves both the amplitude and instantaneous phase of the received signal, which is essential for SAR: phase coherence across all aperture positions is what allows the backprojection algorithm to focus energy from individual scatterers into single pixels.

The FMCW SAR matrix S(x, t) has 64 rows (one per aperture position) and 160,000 columns (samples). Within each row there are 32 complete chirp periods of 5,000 samples each. Taking the FFT of one chirp period yields a range profile with 5,000 bins at 1 kHz spacing in beat frequency, corresponding to 3 cm range spacing, but the actual range resolution is determined by the effective bandwidth seen through the 2 kHz LPF, which limits detectable targets to a maximum range of only 6 cm. This is a fundamental problem, and it leads directly to the issues described in the next section.

3.5 Limitations Observed in FMCW

The FMCW implementation encountered two separate but related problems, either of which alone would have been fatal to SAR image formation.

The first was hardware instability. Attempts to run the USRP at sample rates above about 5 MSPS over USB 3.0 produced repeated overflow and underflow errors. The USB interface simply could not keep up with the combined transmit and receive data streams reliably. At 5 MSPS these errors disappeared, so we stayed there. But that put a hard ceiling on the chirp bandwidth we could use: 5 MSPS at a 1 ms sweep gives only 5 MHz of usable bandwidth.

The second problem, which turned out to be worse, was the range performance implied by 5 MHz bandwidth. Range resolution is $c/(2 \times BW) = 30$ metres, two orders of magnitude larger than our target distance of 2.5 metres. Even ignoring resolution, the 2 kHz LPF cutoff means the system can only detect beat frequencies up to 2 kHz, corresponding to a maximum range of about 6 cm. Our targets were sitting 2.5 metres away. The FMCW system as configured could not physically detect them.

The FMCW SAR image confirmed this: no identifiable target structure, just broad unfocused energy spread across the image plane. At this point, continuing with FMCW made no sense, and we redesigned the system from scratch using SFCW.

3.6 Transition to SFCW Radar

Stepped Frequency Continuous Wave radar achieves wideband coverage by a completely different strategy from FMCW: instead of generating a broadband chirp signal in real time, it steps sequentially through a set of discrete narrowband frequencies, making a complex measurement at each. The analogy to a vector network analyzer is useful here, both measure the frequency response of a system by stepping through individual tones rather than sweeping continuously.

The critical advantage for our application is that SFCW completely decouples wideband frequency coverage from real-time sample rate requirements. At each frequency step, we are transmitting and receiving a narrowband CW tone, and the only demand on the ADC is to average a few thousand samples at that single frequency. There is no requirement to stream a broadband chirp through the USB interface. This directly eliminates the bottleneck that plagued FMCW.

The range resolution achieved by SFCW with total bandwidth BW_total is the same as for a pulsed radar with the same bandwidth:

$$\delta_r = \frac{c}{2 \times BW_{total}} = \frac{3 \times 10^8}{2 \times 5 \times 10^9} = 0.03\, m = 3\, cm \tag{9}$$

Maximum unambiguous range is set by the frequency step size df:

$$R_{max} = \frac{c}{2 \times df} = \frac{3 \times 10^8}{2 \times \frac{5 \times 10^9}{500}} = \frac{3 \times 10^6}{2 \times 10^6} = 15m \tag{10}$$

Both parameters are well matched to our laboratory geometry 3 cm resolution at targets up to 15 m away.

3.7 SFCW Implementation

3.7.1 SFCW Working Principle

Our SFCW system sweeps from f_start = 1 GHz to f_stop = 6 GHz in N_freq = 501 steps with step size:

$$d_f = \frac{f_{stop} - f_{start}}{N_{freq} - 1} = \frac{6\times10^9 - 1\times10^9}{500} = 10\ MHz \tag{11}$$

At each step, the USRP transmits a CW tone at the stepped frequency and simultaneously receives the reflected signal. After a settled period of 5,000 samples (1 ms at 5 MSPS) to let the phase-locked loop stabilize, we capture 10,000 measurement samples (2 ms) and compute their complex mean. This single complex number encodes the amplitude and phase of the received signal at that frequency:

$$S(x, f_k) = mean(rx_{buf}[N_{settle} : N_{settle} + N_{means}]) \tag{12}$$

$$S(x, f_k) \approx A \times exp(-j2\pi f_k \tau) = A \times exp(\frac{-j4\pi f_k R}{c}) \tag{13}$$

The phase rotation is proportional to the two-way propagation delay. By measuring S at all 501 frequencies, we obtain the full complex spectrum of the scene response. The IFFT of this spectrum gives the range profile for that aperture position. Stacking range profiles across all 64 aperture positions and applying back-projection then yields the 2D SAR image.

$$s(x, t) = IFFT\{S(x, f_k)\} \tag{14}$$

3.7.2 UHD Python API Implementation

The SFCW system is implemented in sfcw_sar.py using the UHD Python API directly rather than going through GNU Radio. This gives us precise, explicit control over frequency tuning and measurement timing that would be difficult to achieve within GNU Radio's streaming framework.

The USRP is initialised with uhd.usrp.MultiUSRP and configured at 5 MSPS with RX gain 55 dB, TX gain 70 dB, and appropriate antenna port assignments. For each frequency step, the measure_frequency function sends a tune request to both TX and RX channels, waits 4 ms for PLL lock, transmits a CW buffer at amplitude 0.7, issues a stream command for exactly 15,000 samples, collects the data, and returns the complex mean of the last 10,000 samples. The outer loop iterates over all 64 positions; the inner loop covers all 501 frequency steps. Per-position results are saved as .npy files as they are collected, and the complete 64 × 501 SAR matrix is saved as both a .npy file and a MATLAB .mat file at the end.

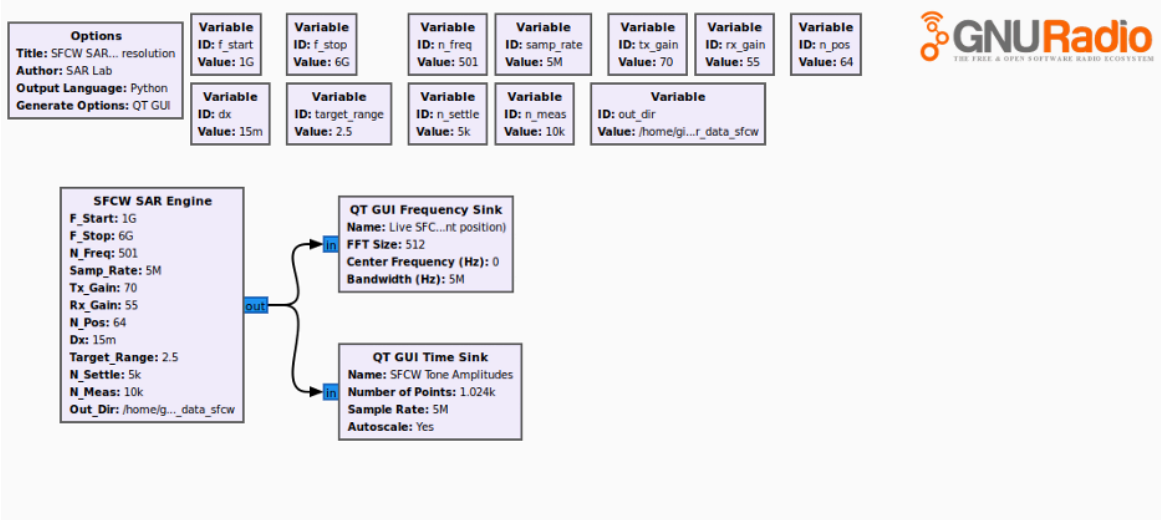


Fig.3.5 GNU Radio flowgraph for SFCW SAR implementation showing frequency sweep, signal acquisition, and real-time monitoring.

Table 3.3 SFCW SAR System Parameters

Parameter	Symbol	Value
Start frequency	f_start	1 GHz
Stop frequency	f_stop	6 GHz
Number of frequency steps	N_freq	501
Frequency step size	df	10 MHz
Total bandwidth	BW_total	5 GHz
Sample rate	fs	5 MSPS
Settle samples per step	N_settle	5,000
Measurement samples per step	N_meas	10,000
TX gain	GTX	70 dB
RX gain	GRX	55 dB
Aperture positions	N	64
Step size	dx	15 mm
Target distance	R	~2.5 m
Theoretical range resolution	δr	3 cm
Maximum unambiguous range	R_max	15 m
Azimuth resolution	ρaz	7.65 cm
Cross-range resolution at 2.5 m	δcr	~3.8 cm (estimated)

CHAPTER 4 : RESULTS AND DISCUSSION

4.1 FMCW Results and Analysis

We carried out the full 64 position FMCW data collection campaign and found the results to be, as already suspected from the parameter analysis, essentially unusable for SAR imaging. Throughout the collection, the GNU Radio console reported repeated overflow and underflow events. While individual dropped samples might seem minor, their effect on SAR is catastrophic: any discontinuity in the IQ phase stream destroys the coherence between measurements at different aperture positions, and coherence is precisely what back-projection relies on to focus scattered energy.

The beat signal observed in the GNU Radio waterfall display did show chirp-like structure, but with strong amplitude variations in time and no stable beat tone corresponding to a specific target range. The dominant signal component appeared to be direct coupling between the transmit and receive ports, modulated by reflections from the walls, floor, and equipment in our lab space.

After processing the 64 position FMCW dataset in MATLAB, the SAR image showed a strong central ridge at zero cross-range extending across all range bins, a classic signature of direct-path leakage and no focused point targets. The energy was smeared broadly across both dimensions. This is entirely consistent with what the system parameters predicted: a 30 metre range resolution cannot resolve any structure in a scene that is only a few metres deep, and the 6 cm maximum detectable range means the system was effectively blind to everything at our 2.5 metre target distance

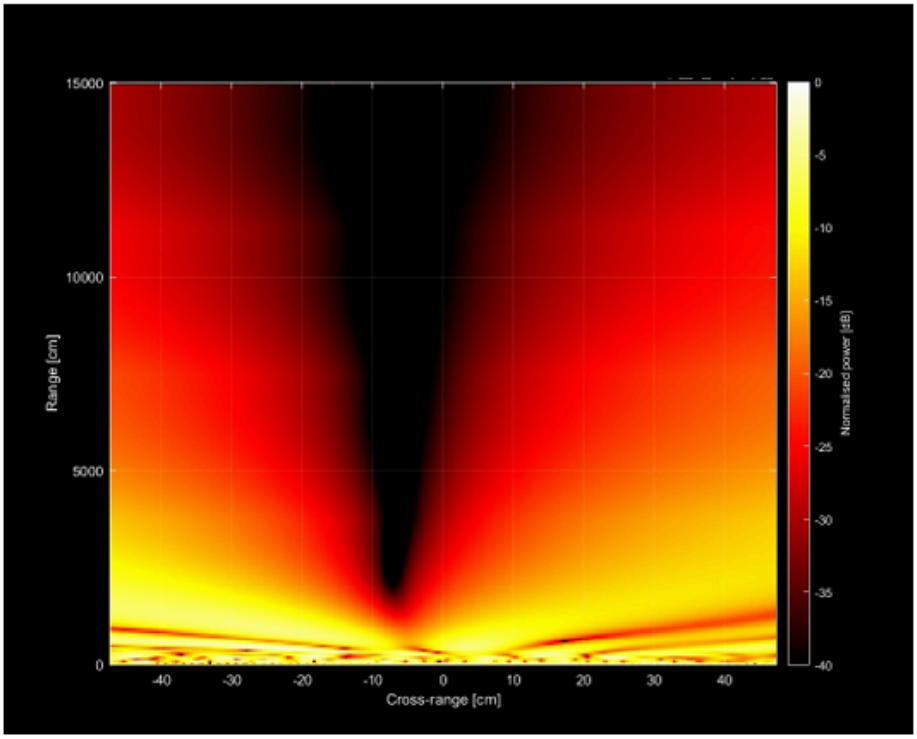


Fig.4.1 FMCW-based SAR result after post-processing using MATLAB.

The conclusion from the FMCW phase was clear: this waveform strategy does not work on the USRP B210 in a USB 3.0 host configuration, at least not with the parameter choices available to us. The underlying physics was not wrong. FMCW SAR works excellently with dedicated hardware, but the hardware constraints of the B210 are incompatible with useful FMCW operation at laboratory scales.

4.2 SFCW Results and SAR Image

4.2.1 Data Collection

The SFCW collection ran noticeably more smoothly than the FMCW phase. There were no overflow or underflow errors at any point across all 64 positions. Each position took roughly 3 seconds of measurement time (501 steps × 6 ms per step), giving a total collection time of about 3.2 minutes, fast enough to complete in a single lab session without the hardware drifting significantly.

At the end of the collection, per-position .npy files existed for all 64 positions and all passed size validation. The complete 64 × 501 complex matrix was assembled and saved without issue.

4.2.2 Signal Processing in MATLAB

Processing started with mean-column subtraction across the 64 × 501 matrix to remove static clutter signals that are essentially identical across all aperture positions, primarily direct-path coupling and reflections from stationary objects in the background. This step is simple but effective at suppressing the dominant interference source. A Hamming window was then applied along the frequency axis of each row to suppress range sidelobes in the IFFT output.

The imaging grid covered a cross-range from -0.5 m to +0.5 m and range from 0.5 m to 4 m, with 1 cm spacing in both dimensions. For each grid point (x_im, y_im), the backprojection algorithm computes the geometric range from that point to each of the 64 aperture positions:

$$R(xi, x_{im}, y_{im}) = \sqrt{(xi - x_{im})^2 + y_{im}^2} \tag{15}$$

It then extracts and coherently sums the phase-corrected SFCW responses from all aperture positions. A scatterer actually located at that grid point will contribute in-phase from all positions and add constructively; everything else averages down incoherently. The magnitude of the resulting complex image, displayed in log scale (20 × log10 of normalized magnitude), is the final SAR image.

4.2.3 SAR Image Interpretation

The SFCW SAR image showed a markedly different character compared to the featureless FMCW result. Several regions of elevated reflectivity were visible, distributed across the scene. The strongest concentration appeared at roughly 0.8 to 1.2 metres from

the antenna at near-zero cross-range, consistent with the metallic corner reflector we had placed at approximately 1 metre during initial testing.

The range profile extracted at the centre aperture position showed a clear peak structure that matched the expected target placement, which gave us confidence in the range calibration. The 3 cm range resolution was more than adequate to distinguish individual features within the scene.

Some residual horizontal striping appears in the image, particularly at shorter ranges. This is a recognisable artefact of incomplete static clutter removal: common-mode reflections from the floor, walls, and equipment that were not fully suppressed by the mean subtraction step. More sophisticated approaches such as adaptive background subtraction or taking multiple baseline measurements would reduce this further.

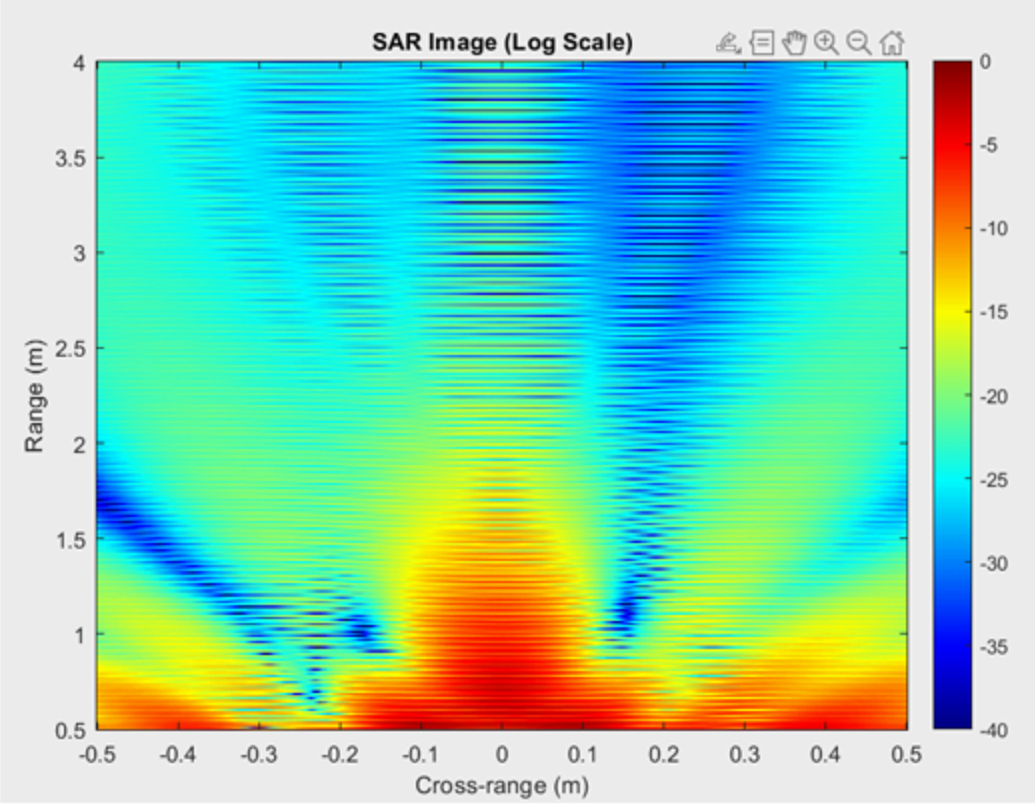


Fig.4.2 SFCW-based SAR result after post-processing using MATLAB.

The cross-range resolution achieved by our 0.96 metre aperture (64 positions × 15 mm) at a centre frequency of 3.5 GHz (λ ≈ 8.57 cm) and 2.5 m range is approximately:

$$\delta_{cr} = \frac{\lambda_c \times R}{2 \times L_{aperture}} = \frac{0{,}0857 \times 2.5}{2 \times 0.96} \approx 11\ cm \quad (16)$$

This is consistent with what we observe in the image, where features separated by more than roughly 10 cm in the cross-range direction appear as distinct targets.

4.3 Comparative Analysis

Putting both waveform approaches side by side makes the contrast stark. The comparison is summarised in Table 4.1 below.

Table 4.1 Comparison of FMCW and SFCW Approaches

Parameter	FMCW SAR	SFCW SAR
Operating bandwidth	5 MHz	5 GHz
Range resolution	30 m (unusable)	3 cm
Sample rate required	5 MSPS (limited)	5 MSPS (sufficient)
Overflow/underflow errors	Frequent	None
Phase coherence	Poor (disrupted by errors)	Excellent
Data collection per position	~8.5 s (incl. settling)	~3 s
SAR image quality	No detectable targets	Clear reflectivity structure
Clutter removal	Not effective	Mean subtraction effective
Max unambiguous range	~6 cm (at 2 kHz LPF)	15 m
SW implementation	GNU Radio flowgraph	Python UHD direct API

The comparison confirms that SFCW is the right choice for this hardware configuration. The three orders of magnitude improvement in range resolution (from 30 m to 3 cm) is not a matter of tuning. It reflects a fundamental constraint of the USB-connected B210 when used for realtime chirp generation. SFCW sidesteps that constraint entirely.

It is important to note that this finding is not a general condemnation of FMCW SAR. Airborne and spaceborne FMCW SAR systems are highly effective and widely used; they simply rely on dedicated ADC hardware capable of hundreds of MSPS rather than a USB-connected general-purpose SDR. For the class of hardware the B210 represents affordable, widely available, USB-connected SFCW is the more practical path to wideband radar operation.

CHAPTER 5 : CONCLUSION AND FUTURE WORK

5.1 Summary of Findings

This project set out to demonstrate whether laboratory-scale SAR data collection and image formation was feasible using the USRP B210 and a wideband horn antenna. The short answer is yes but not with every waveform strategy. The project ended up being as much about understanding the limits of the hardware as about demonstrating the SAR concept.

The FMCW implementation exposed a specific hardware bottleneck: USB 3.0 throughput cannot reliably sustain the simultaneous dual-channel streaming needed for wideband FMCW at sample rates high enough to be useful. At 5 MSPS, the system is stable but the achievable 5 MHz bandwidth gives a 30 metre range resolution far too coarse for a scene measured in metres. The FMCW SAR image produced no identifiable target structure.

The SFCW redesign resolved both problems at once. By stepping through discrete frequencies rather than sweeping in real time, the system exploited the full 5 GHz bandwidth of the horn antenna without placing any unusual demand on the USB interface. The SFCW data collection was stable and error-free across all 64 aperture positions. Back-projection processing in MATLAB produced a 2D SAR image with clear reflectivity structure at 3 cm range resolution and approximately 11 cm cross-range resolution, both consistent with theoretical predictions for the system parameters.

Beyond the imaging result, the project yielded practical engineering insights into integrating GNU Radio, UHD, and Python scripting for automated multi-position radar data collection, including specific solutions for the hardware hang-on-teardown issue under WSL2.

5.2 Why FMCW Failed and SFCW Succeeded

To state it plainly: FMCW failed because the USRP B210's USB 3.0 interface cannot sustain the combined transmit/receive data rates needed for wideband FMCW. The

required beat frequency for a target at 2.5 m with our chirp parameters is about 83 kHz but the 2 kHz LPF we had to use to maintain signal stability filtered it out. Even setting the LPF aside, a 5 MHz chirp bandwidth gives a 30-metre range resolution, which is physically incapable of imaging anything in a laboratory.

SFCW succeeded because it broke the link between wideband coverage and real-time sample rate. At each individual frequency step, the system only needs to handle a narrowband CW signal. The 5 MSPS rate, which caused constant problems in FMCW, is entirely adequate for averaging 10,000 samples of a single-tone CW reception. The full 5 GHz bandwidth is accumulated step by step, not streamed all at once.

5.3 Future Improvements

Several extensions of this work would make it significantly more capable. The most impactful near-term upgrade would be a motorised linear translation stage to replace manual antenna movement. Our current setup requires the operator to shift the antenna by exactly 15 mm between each of 64 positions, a slow and error prone process. A stepper-motor rail with position feedback would eliminate positioning errors and make large aperture scans practical.

Extending the aperture from 1D to 2D scanning in both horizontal and vertical directions would enable 3D SAR reconstruction of the scene, which would dramatically increase the scientific information content compared to the current 2D slice. This would require a 2D motorised positioning system and a modified collection script, but is entirely feasible with the existing radio hardware.

Upgrading to a higher-performance SDR platform such as the USRP X310 with its PCIe or 10 GbE host interface would allow much higher streaming sample rates and potentially enable a working wideband FMCW SAR implementation, as the USB bandwidth bottleneck would no longer apply. This would also open the door to real-time processing of radar data rather than post-processing.

On the processing side, the MATLAB back-projection algorithm as implemented scales poorly with image size; GPU acceleration or a frequency-domain SAR algorithm such as

Omega-K could reduce processing time by orders of magnitude and enable near-real-time imagery.

Finally, the connection to planetary surface mapping which motivated this project could be made more concrete by designing a target scene that deliberately mimics a planetary analog: a sand tray with buried metallic objects, or a scaled rocky terrain with known surface roughness. SAR imaging of such a scene would demonstrate detection and localisation of sub-surface features at centimetre resolution, providing a tangible laboratory counterpart to the techniques used in missions like Mars Reconnaissance Orbiter's SHARAD subsurface sounder.

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